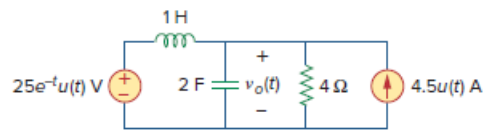


Problem

Find $v_o(t)$ in the circuit of Fig. 16.58.

Figure 16.58:



Step-by-step solution

Step 1 of 5

Refer to Figure 16.58 for the circuit in the textbook.

For $t < 0$, note that the voltage and current sources turn OFF.

Since,

$$u(t) = \begin{cases} 1, & t \geq 0 \\ 0, & t < 0 \end{cases}$$

Consider the following initial conditions for $t < 0$.

$$i(0) = 0\text{ A}$$

$$v(0) = 0\text{ V}$$

Step 2 of 5

Transform the circuit from time domain to s -domain

Convert the elements from time domain to s -domain

Apply the Laplace Transform for the following elements in the circuit

Laplace of unit-step function is:

$$L[u(t)] = \frac{1}{s}$$

For the input voltage element in the circuit shown in figure (1) transform the input voltage element from the time domain to s -domain is:

$$L[25e^{-t}u(t)] = 25L[e^{-t}u(t)]$$

$$L[25e^{-t}u(t)] = \frac{25}{s+1}$$

For the output current element in the circuit shown in figure (1) transform the output current element from the time domain to s -domain is:

$$L[4.5u(t)] = 4.5L[u(t)]$$

$$L[4.5u(t)] = \frac{4.5}{s}$$

For capacitor element of 2 F :

$$\frac{1}{sC} = \frac{1}{s \cdot 2}$$

$$\frac{1}{sC} = \frac{0.5}{s}$$

For Inductor element of 1 H :

$$sL = s \cdot 1$$

$$sL = s$$

Step 3 of 5

Draw s -domain circuit as shown in figure (2).

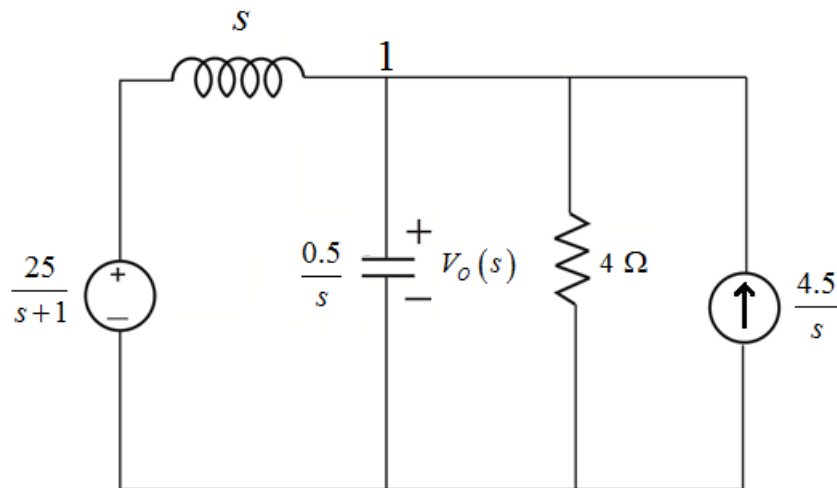


Figure (2)

Step 4 of 5

Apply the source transformation to the circuit shown in figure (2).

The equivalent value of current source $\frac{4.5}{s}$ Amperes is in parallel with 4ohms resistor is,

$$V_{eq} = \frac{4.5}{s} \times 4$$

$$V_{eq} = \frac{18}{s}$$

Replace the voltage source $\frac{18}{s}$ Volts is in series with 4ohms resistor which is connected at the output terminal of the circuit as shown in figure (3).

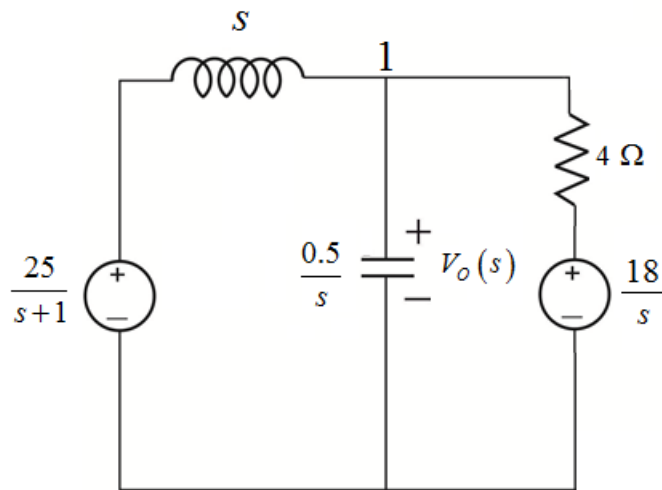


Figure (3)

Apply the Kirchhoff's current law at node 1 of the circuit shown in figure (3).

$$\frac{V_o(s) - \frac{25}{s+1}}{s} + \frac{V_o(s)}{\left(\frac{0.5}{s}\right)} + \frac{V_o(s) - \frac{18}{s}}{4} = 0$$

$$V_o(s) \left(\frac{1}{s} + 2s + \frac{1}{4} \right) = \frac{25}{s(s+1)} + \frac{18}{4s}$$

$$V_o(s) \left(\frac{4+8s^2+s}{4s} \right) = \frac{100+18(s+1)}{4s(s+1)}$$

$$V_o(s) = \frac{118+18s}{(s+1)(8s^2+s+4)}$$

Simplify further,

$$\begin{aligned} V_o(s) &= \frac{2(400s-449)}{11(8s^2+s+4)} + \frac{100}{11(s+1)} \\ &= \frac{800s-898}{88 \left(s^2 + \frac{1}{8}s + \frac{1}{2} \right)} + \frac{100}{11(s+1)} \\ &= \frac{800s-898}{88 \left[s^2 + 2 \left(\frac{1}{16} \right) s + \left(\frac{1}{16} \right)^2 + \left\{ \frac{1}{2} - \left(\frac{1}{16} \right)^2 \right\} \right]} + \frac{100}{11(s+1)} \\ &= \frac{800s-898}{88 \left[\left(s + \frac{1}{16} \right)^2 + \left(\frac{\sqrt{127}}{16} \right)^2 \right]} + \frac{100}{11(s+1)} \\ &= \frac{800s}{88 \left[\left(s + \frac{1}{16} \right)^2 + \left(\frac{\sqrt{127}}{16} \right)^2 \right]} - \frac{898 \left(\frac{16}{\sqrt{127}} \right) \left(\frac{\sqrt{127}}{16} \right)}{88 \left[\left(s + \frac{1}{16} \right)^2 + \left(\frac{\sqrt{127}}{16} \right)^2 \right]} + \frac{100}{11(s+1)} \\ &= \frac{800s}{88 \left[\left(s + \frac{1}{16} \right)^2 + \left(\frac{\sqrt{127}}{16} \right)^2 \right]} - \frac{898 \left(\frac{16}{\sqrt{127}} \right) \left(\frac{\sqrt{127}}{16} \right)}{88 \left[\left(s + \frac{1}{16} \right)^2 + \left(\frac{\sqrt{127}}{16} \right)^2 \right]} + \frac{100}{11(s+1)} \end{aligned}$$

Apply the inverse Laplace transform.

$$\begin{aligned} v_o(t) &= 9.091e^{\frac{-1}{16}t} \cos\left(\frac{\sqrt{127}}{16}t\right) u(t) - e^{\frac{-1}{16}t} 14.488 \sin\left(\frac{\sqrt{127}}{16}t\right) u(t) + 9.091e^{-t} u(t) \\ &= 9.091e^{-0.0625t} \cos\left(\frac{\sqrt{127}}{16}t\right) u(t) - e^{-0.0625t} 14.488 \sin\left(\frac{\sqrt{127}}{16}t\right) u(t) + 9.091e^{-t} u(t) \\ &= e^{-0.0625t} \left[9.091 \cos(0.7044t) - 14.488 \sin(0.7044t) \right] u(t) + 9.091e^{-t} u(t) \\ &= e^{-0.0625t} \left[\frac{9.091}{(-0.4625)} \cos(0.7044t) \cos(117.55^\circ) - \frac{14.488}{0.8866} \sin(0.7044t) \sin(117.55^\circ) \right] u(t) + 9.091e^{-t} u(t) \\ &= \left[9.091e^{-t} + 19.653e^{-0.0625t} \cos(0.7044t - 117.55^\circ) \right] u(t) \text{ V} \end{aligned}$$

Therefore, the voltage across the capacitor is,

$$\boxed{\left[9.091e^{-t} + 19.653e^{-0.0625t} \cos(0.7044t - 117.55^\circ) \right] u(t) \text{ V} .}$$